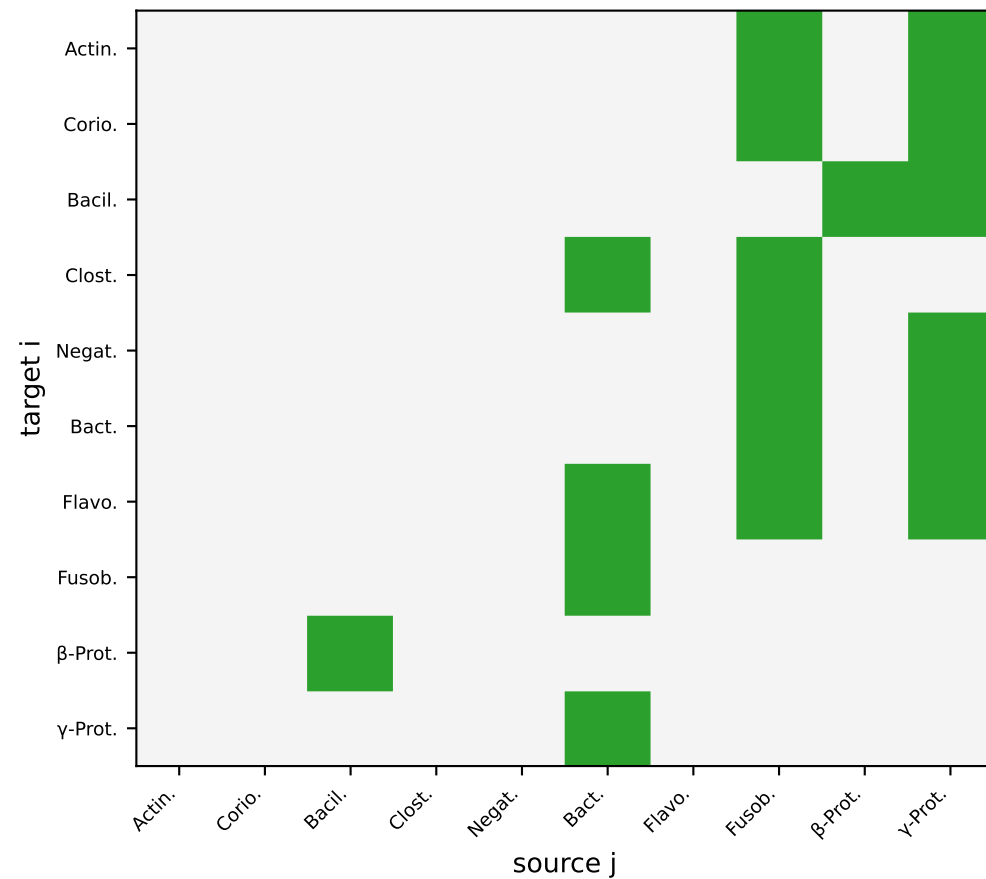


Guild-level sign validation: manually curated minimal metabolic models
(HOMD + Kolenbrander 2010 literature basis; oral-fluid pFBA)

A. Minimal-model sign predictions
(green=+, red=−, grey=uncertain)



B. Cross-feeding score
 $\log(1 + \sum \text{importable flux, } j \rightarrow i)$

